# Transformers and ChatGPT KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

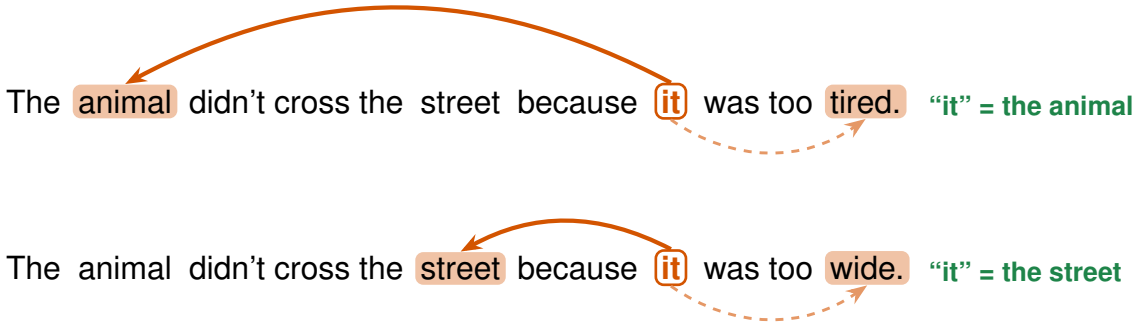
## 1. What does "it" mean?

The animal didn't cross the street because **it** was too **tired**. "it" means the animal

The animal didn't cross the street because **it** was too **wide**. "it" means the street

Which word told you? the last word: tired or wide

## 2. Attention.




A transformer looks at every word. For each word, it decides which other words matter most.

This is called attention. Google scientists shared the transformer in 2017.

## 3. GPT and ChatGPT.

**G** = Generative — it makes new text

**P** = Pre-trained — it learned first from a huge amount of text

**T** = Transformer — it uses attention to connect the words

A chatbot writes by predicting the next word, again and again.

ChatGPT came out in November 2022. About 100 million people used it in about 2 months.

In 2024, two Nobel Prizes went to scientists who worked on AI.

**4. Be the chatbot: next word, by vote.**

| Word | The word with the most votes | How many votes? |
|---|---|---|
| 1 | Answers vary. | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |

Our class sentence: After school, my friends and I Answers vary.

The class AI tool wrote: Answers vary.

Was the class sentence **surprising** or **ordinary**? Why? Ordinary. The most common word wins.

**5. The class timeline.** Number the moments from 1 (first) to 6 (last).

| | | | |
|---|---|---|---|
| 4 | ImageNet: 14 million labeled pictures | 1 | Alan Turing asks: Can a machine think? |
| 6 | ChatGPT | 3 | Deep Blue beats the world chess champion |
| 2 | ELIZA, one of the first chatbots | 5 | Transformers and attention |

**6. Talk about it.** ChatGPT learned from text that people wrote. Most writers were never asked.

Writers **should** / **should not** be asked first, because Answers vary.

*✂ Cut out the cards. Add them to the class timeline.*

**2017**

Google scientists share the **transformer**. It uses attention.

Draw a picture here.

**2022**

**ChatGPT** comes out. About 100 million people use it in 2 months.

Draw a picture here.

**2024**

Scientists who worked on AI win two **Nobel Prizes**.

Draw a picture here.